CONCEPT OF ACTIVITY AND ACTIVITY COEFFICIENT.

In order to describe correctly the properties and behaviour of a non-ideal mixture or solution the "strength", i.e the mole fraction, molality, concentration or partial pressure of each substance must be replaced by the activity of the substance in the appropriate formulae. The correction factor which is used to correct the 'strength' so as to give the activity is known as the activity coefficient. i.e

activity = activity coefficient × 'strength.'

G. N. Lewis discovered that the experimentally determined value of concentration whether of molecules or ions in solution is less than the actual concentration. The apparent value of concentration is termed 'activity'. It may be defined as the effective concentration of a molecule or ion in solution.

The activity coefficient 'γ' is defined as the ratio between the activity, denoted by a, or effective concentration and actual concentration of the molecule or ion in solution that is:-

$$\gamma = \frac{\text{effective concentration}}{\text{actual concentration}} = \frac{a}{c}.$$

or $\boxed{a = \gamma c}$.

γ can be determined experimentally and is given in tables. Thus the value of activity can be calculated by applying the above relation.

<u>NB</u>: The effective number of cations and anions in solutions becomes less as these tend to form ion-pairs because of strong electrostatic attraction. Thus, the effective concentration of the ion becomes less than the actual concentration.

- The activity coefficient, γ of strong electrolytes is always less than 1.
- The value of γ decreases with dilution at the same temperature, and at infinite dilution approaches 1.
- In the mathematical expressions of the various laws of physical chemistry such as Raoult's law, Henry's law, law of mass action and Ostwald's law the actual concentration (c) has to be replaced with effective concentration (a) for accurate experimental work

Example: Calculate the effective concentration of a 0.0992M solution of NaCl at 25°C for which activity coefficient is 0.782.

Solution;

We know that $a = \gamma c$

On substitution; $a = (0.782)(0.0992M)$
$= 0.0776M$

Arrhenius theory of Ionisation.

Arrhenius studied the conduction of current through water solutions of electrolytes. He came to believe that the conductivity of solutions was due to the presence of ions. In 1884, Arrhenius put forward his theory of ionisation. Arrhenius theory of ionisation may be studied as;

(1) When dissolved in water, neutral electrolyte molecules are split up into two types of charged particles. These particles are called ions and the process was termed ~~dissociation~~ ionisation. The +vely charged particles were called cations and those having -ve charge were called anions.

In the modern form, the theory assumes that the ions are already present in the solid electrolyte and these are held together by electrostatic force.

When applied in water, these neutral molecules dissociate to form separate anions and cations

Thus,

$AB \rightarrow A^+ + B^-$ — — - (old view)

$A^+B^- \longrightarrow A^+ + B^-$ (modern view)

for this reason, the theory may be referred as the theory of electrolytic dissociation.

(2) The ions present in solution constantly reunite to form neutral molecules. Thus there is a state of equilibrium b/w the undissociated molecules and the ions.

$$AB \rightleftharpoons A^+ + B^-.$$

Applying the law of mass action to the ionic equilibrium we have;

$$\frac{[A^+][B^-]}{[AB]} = K$$ K is called Dissociation Constant

(3) The charged ions are free to move through the solution to the oppositely charged electrode. This movement of the ions constitutes the electric current through electrolytes. This explains the conductivity of electrolytes as well as the phenomenon of electrolysis.

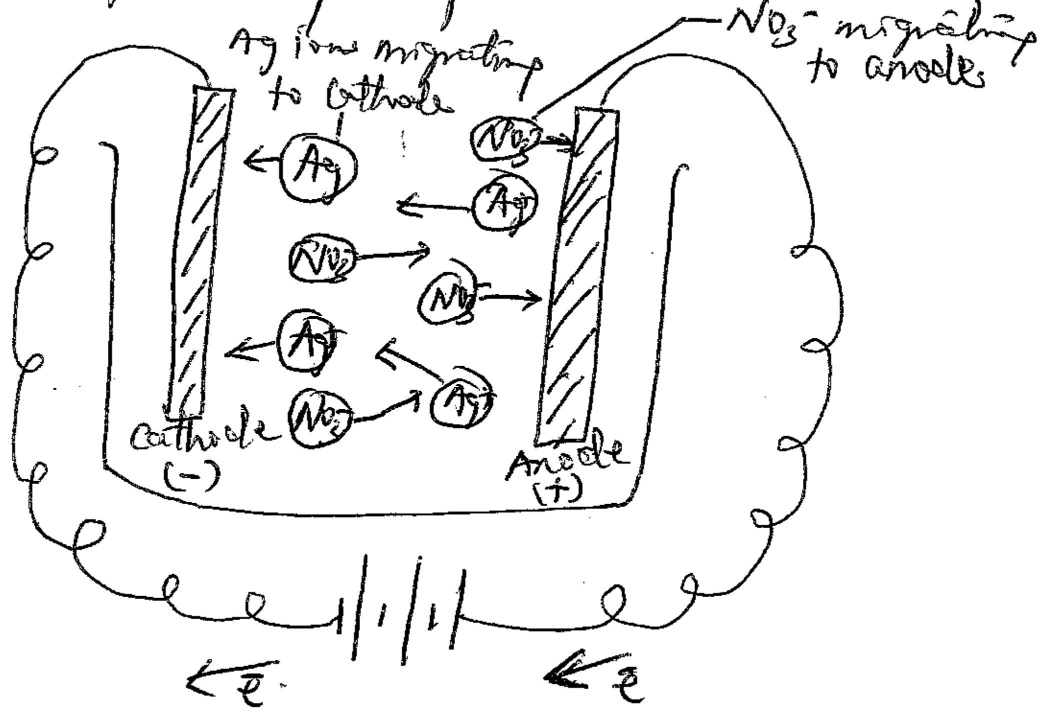
(4) The electrical conductivity of an electrolyte solution depends on the number of ions present in solution. Thus the degree of dissociation of an electrolyte determines whether it is a strong electrolyte or a weak electrolyte.

Migration of Ions.

We know that electrolytes dissociate in solution to form +ve ions and −ve ions.

$$AgNO_3 \longrightarrow Ag^+ + NO_3^-$$
$$CuSO_4 \longrightarrow Cu^{2+} + SO_4^{2-}$$

As the current is passed b/w the electrodes of the electrolytic cell, the ions migrate to the opposite electrodes. Thus in the electrolytic solution of $AgNO_3$, the cations (Ag^+) will move to the cathode and anions (NO_3^-) to the anode. Usually, different ions move with different rates. The migration of ions through the electrolytic solution can be demonstrated by the following experiments.



Migration of ions through electrolytic solution to opposite electrodes

Relative Speed of Ions

It have been Considered that ions move to the oppositely Charged electrodes Under the influence of the electric Current. But the speeds of Cations Migrating towards the Cathode and those of anions migrating towards the anode are not ~~proportional~~ necessarily the same. However, the speed of a Cation moving away from the anode will be proportional to the fall of Concⁿ of these ions at the anode. Similarly, the speed of an anion moving away from the Cathode will be proportional to the fall of Concⁿ of anions around the Cathode. Hittorf Studied such Changes experimentally and gave a general rule known as Hittorf's Rule. It state that "the loss of Concⁿ around any electrode is proportional to the speed of the ion moving away from it.

We can write the expression:

$$\frac{\text{Fall around anode}}{\text{fall around Cathode}} = \frac{\text{Speed of Cation}}{\text{Speed of anion}} = \frac{V_+}{V_-}$$

where V_+ and V_- is the speed of Cations and anions respectively. The above relation apply when the Ions do not react with the material of the electrodes. However, in many cases they combine with the material of the electrodes rather than depositing on it. this results in an increase in Concⁿ around ~~the~~ such an electrode instead of a decrease.

Example: In the expt, the increase in conc^{n} of $AgNO_3$ around the silver anode was 5.6 mg of silver. 10.73 mg of silver were deposited by the same current in the silver coulometer placed in series. Find the speed ratio of Ag^+ and NO_3^- ions.

Solution:

Fall of conc. around cathode (= rise of conc. around anode) = 5.6 mg.

If no Ag^+ ions had migrated from the anode, the increase in conc. around the anode would have been 10.73 mg silver. But the actual increase is 5.6 mg

$\therefore$ fall around the anode due to migration of Ag^+

$$= (10.73 - 5.6) = 5.13.$$

$$\frac{\text{Speed of } Ag^+}{\text{Speed of } NO_3^-} = \frac{\text{fall around anode}}{\text{fall around cathode}} = \frac{5.13}{5.6} = \underline{\underline{0.916}}$$

Transport Number.

During electrolysis, the current is carried by the anions and the cations. The fraction of the total current carried by the cation or the anion is termed its Transport number or Hittorf's number

If V_+ represents the speed of migration of the cation and V_- that of the anions

the transport no of cation $= \dfrac{V_+}{V_+ + V_-}$.

The transport no of anion $= \dfrac{V_-}{V_+ + V_-}$.

The transport no of the Cation is represented by t_+ and that of the anions by t_-.

Thus, $t_+ = \frac{V_+}{V_+ + V_-}$ and $t_- = \frac{V_-}{V_+ + V_-}$ ⊛

or $\frac{t_+}{t_-} = \frac{V_+}{V_-}$ and $t_+ + t_- = 1$ ⊛

If the speed ratio V_+/V_- be denoted by r, we have

$$r = \frac{t_+}{t_-} = \frac{t_+}{1 - t_+}$$

and $$t_- = \frac{1}{1+r}.$$

<u>Example</u>: The speed ratio of silver and nitrate ions in a solution of silver nitrate electrolysed b/w silver electrode is 0.916. Find the transport no of the two ion.

<u>Solution</u>:

We know that $t_- = \frac{1}{1+r}$

where t_- is the transport no of the anions and r is the speed ratio of the anion and the Cation.

$$\therefore t_{NO_3^-} = \frac{1}{1 + 0.916} = 0.522$$

and $$t_{Ag^+} = 1 - t_{NO_3^-} = 1 - 0.522$$
$$= \underline{0.478}$$

The two methods for determination of the transport no of ions are (1) Hittorf's method (2) Moving boundary method.

Kohlrausch's law

from the study of the equivalent conductances of different electrolytes at infinite ~~electrolytes~~ dilution (λ_∞), Kohlrausch's discovered that each ion contributes to the conductance of the solution. In 1875, he enunciated a generalisation which is called the Kohlrausch's law which states that "the equivalent conductance of an electrolyte at infinite dilution is equal to the sum of the equivalent conductances of the component ions." Mathematically, it can be expressed as:

$$\lambda_\infty = \lambda_a + \lambda_c .$$

where λ_a is the equivalent conductance of the anion and λ_c that of the cation.

for example, the equivalent conductance of NaCl at infinite dilution at 25°C is found to be 126.45. The equivalent conductance of Na^+ and Cl^- ion is 50.11 ohm^{-1} and 76.34 ohm^{-1} respectively. Thus:

$$\lambda(NaCl) = \lambda_{Cl^-} + \lambda_{Na^+}$$

or $$126.45 = 50.11 + 76.34$$

this is in conformity with the Kohlrausch's law

Example: Cal. the equivalent conductance at 20°C of NH_4OH at infinite dilution. Given: $\lambda_\infty(NH_4Cl) = 130$, $\lambda_\infty(OH^-) = 174$, $\lambda_\infty(Cl^-) = 66$.

Solution

Applying Kohlrausch's law.

$$\lambda_\infty(NH_4OH) = \lambda_\infty(NH_4Cl) + \lambda_\infty(OH^-) - \lambda_\infty(Cl^-)$$

$$= 130 + 174 - 66$$

$$= 238\ ohm^{-1}cm^{-1}$$

The Ostwald's Dilution law

According to the Arrhenius theory of dissociation, an electrolyte dissociates into ions in water solutions. These ions are in a state of equilibrium with the undissolved molecules. This equilibrium is called the ionic equilibrium. Ostwald noted that the law of Mass Action can be applied to the ionic equilibrium as in the case of chemical equilibria.

Consider a binary electrolyte AB which dissociates in solution to form the ions A^+ and B^-

$$AB \rightleftharpoons A^+ + B^-$$

Let C moles per litre be the conc. of the electrolyte and α (alpha) its degree of dissociation. The conc. terms at equilibrium may be written as:-

$$[AB] = C(1-\alpha)\ mol\,L^{-1}$$

$$[A^+] = C\alpha\ mol\,L^{-1}$$

$$[B^-] = C\alpha\ mol\,L^{-1}$$

Applying the law of Mass Action:

Rate of dissociation $= k_1 \times C(1-\alpha)$

Rate of Combination $= k_2 \times C\alpha \times C\alpha$

At equilibrium: $k_1 \times C(1-\alpha) = k_2 \times C\alpha \times C\alpha$

or $$\frac{C\alpha \times C\alpha}{C(1-\alpha)} = \frac{k_1}{k_2} = K_c$$

or $$K_c = \frac{\alpha^2 C}{(1-\alpha)} \text{ mol l}^{-1} \quad \text{——— (1)}$$

The equilibrium Constant K_c is called the <u>Dissociation Constant or Ionisation Constant</u>. It has a Constant value at a Constant temperature.

If one mole of an electrolyte be dissolved in V litre of the solution, then

$$C = \frac{1}{V}$$

V is known as the Dilution or solution.

Thus expression (1) becomes,

$$K_c = \frac{\alpha^2}{(1-\alpha)V} \quad \text{- - - - (2)}$$

this expression which correlates the variation of the degree of dissociation of an electrolytes with dilution, is known as <u>Ostwald's Dilution Law</u>

<u>For Weak Electrolytes</u>

the value of α is very small as compared to 1, so that in most of the Calculations we can take $1-\alpha \simeq 1$.

Thus the Ostwald's Dilution Law expression becomes

$$K_c = \frac{\alpha^2}{V} \quad --- \quad (3)$$

It implies ~~that~~ here that the degree of dissociation of a weak electrolyte is proportional to the square root of the dilution i.e.

$$\alpha \propto \sqrt{K_c V}$$

$$\text{or} \quad \alpha = K' \sqrt{V}$$

For Strong Electrolytes

for strong electrolytes, the value of α is large and it cannot be neglected in comparison with 1. thus we have to use the original expression (2), i.e.

$$K_c = \frac{\alpha^2}{(1-\alpha)V}$$

$$\text{or} \quad \alpha^2 = K_c (1-\alpha) V = K_c V - \alpha K_c V.$$

which gives a quadratic equation:

$$\alpha^2 + \alpha K_c V - K_c V = 0. \quad --- \quad (4).$$

Limitation of Ostwald's Law

The law holds good only for weak electrolytes and fails completely when applied to strong electrolytes. for strong electrolytes, which are highly ionised in solution, the value of the dissociation constant K, far from remaining constant, rapidly falls with dilution.

THEORY OF STRONG ELECTROLYTES

A number of theories have been put forward by different workers in order to explain the high conductance of strong electrolytes. Southerland (1906) held the view that ions in solution were surrounded by a large no of ions of opposite charge. Due to the weakening of interionic forces, the ionic velocities were accelerated. This resulted in the increase of conductance of the electrolytes solution. However, not much notice was taken of Southerland's view.

Ghosh's Formula:

In 1918 J.C. Ghosh reviewed the above theory. He assumed that though the electrolyte is completely ionised, all the ions are free to move owing to the influence of the electric charges and it is only the mobile ions which contribute to the conductance of the solution. The value α represents the "ctive" proportion of the electrolyte and can be determined by purely electrical data, the law of Mass Action playing no part whatsoever. His formula:

$$\boxed{\sqrt[3]{V}\log\alpha = K.}$$

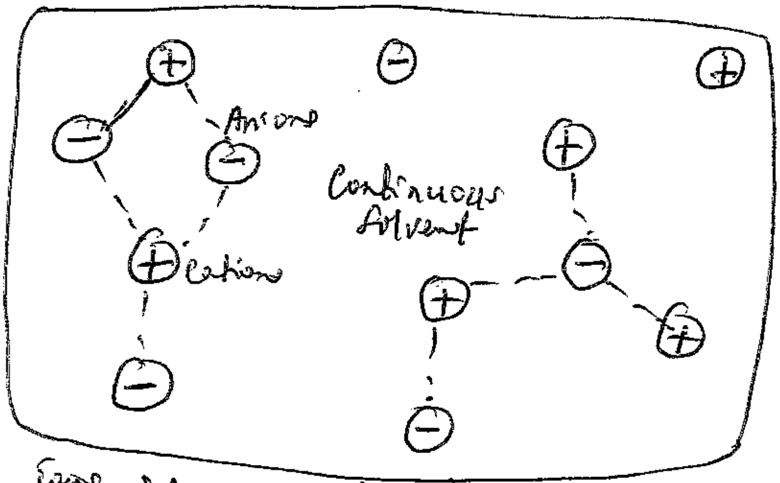
was applicable to univalent strong electrolytes.

Debye-Huckel Theory

In 1923 Debye and Huckel and in 1926 Onsager put forward the modern theory of strong electrolytes in which account is taken of the electrostatic forces b/w the ions. A brief outline of the main ideas of the theory is given as:-

(1) The strong electrolyte is completely ionised at all dilutions.

(2) Since oppositely charged ions attract each other, it suggests that anions and cations are not uniformly distributed in the solution of an electrolyte but that the cations tend to be found in the vicinity of anions and vice-versa as shown below.



The ions of a particular charge are surrounded by more ions of the opposite and solvent molecules. Because of the large number of ions in concentrated solution, the ion activity is reduced owing to hindered movement of the ions.

(3) Decrease in equivalent conductance with increase in concentration is due to fall in mobilities of the ions due to greater inter-ionic effect and vice-versa.

(4) The ratio $\lambda_v / \lambda_\infty$ does not correctly give the degree of dissociation α for strong electrolytes but only the conductance or conductance coefficient f_c.

(5) In spite of almost complete ionisation, λ_v is much less than λ_∞.

<u>Debye-Huckel-Onsagar Conductance Equation</u>: takes these causes into account and for a uni-univalent electrolyte supposed to be completely dissociated is written in the form:

$$\lambda_v = \lambda_\infty - (A + B\lambda_\infty)\sqrt{c}.$$

where A and B are constants and c is the concn in gm-equivalents per litre. These constants depend only on the nature of the solvent and the temperature and are given by the relationships:

$$A = \frac{82.4}{(DT)^{1/2}\eta} \quad \text{and} \quad B = \frac{8.20\times10^{5}}{(DT)^{3/2}}$$

where D and η are the dielectric constant and coefficient of viscosity of the medium respectively at the absolute temperature T. The constant A is a measure of the electrophoretic effect while B is that of the ~~asymmetric~~ asymmetry effect.

For example, for water at 25°C with $D = 78.5$ and $\eta\times10^{3} = 8.95$, the value of A is 60.20 and that of B is 0.229. On substituting these values in the above equation, we have,

$$\lambda_v = \lambda_0 - (60.20 + 0.229\lambda_0)\sqrt{C}$$

It follows therefore that, if the above equation is correct a straight line of slope equal to $(60.20 + 0.229\lambda_0)$ should be obtained by plotting observed equivalent conductance (λ_v) against the square root of the corresponding concentration i.e, $\sqrt{C}$.

Degree of Dissociation

When a certain amount of electrolyte (A^+B^-) is dissolved in water, a small fraction of it dissociates to form ions (A^+ and B^-). When the equilibrium has been reached b/w the undissociated and the free ions, we have

$$AB \rightleftharpoons A^+ + B^-.$$

The fraction of the amount of the electrolyte in solution present as free ions is called the <u>Degree of dissociation.</u>

If the degree of dissociation is represented by x, we can write,

$$x = \frac{\text{amount dissociated (mol/L)}}{\text{Initial Concentration (mol/L)}}$$

The value of x can be calculated by applying the Law of Mass Action to the ionic equilibrium stated above:

$$K = \frac{[A^+][B^-]}{[AB]}$$ If K is known, x can be calculated.

The Common-ion Effect

When a soluble salt (say A^+C^-) is added to a solution of another salt (A^+B^-) containing a common ion (A^+), the dissociation of AB is suppressed.

$$AB \rightleftharpoons A^+ + B^-$$

By the addition of the salt (AC), the concentration of A^+ increases. Therefore, according to Le Chatelier's principle, the equilibrium will shift to the left, thereby decreasing the concentration of A^+ ions. Or that, the degree of dissociation of AB will be reduced.

The reduction of the degree of dissociation of salt by the addition of a common ion is called the <u>Common-ion effect</u>

Application of the Common-ion effect

(1) In a saturated solution of silver chloride, we have the ⇌brm.

$$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$$

When sodium chloride is added to the solution, the concentration of Cl^- ions will increase. The ⇌brm shown above will be shifted to the left to form more of solid AgCl. Thus the solubility of AgCl, a typical sparingly soluble salt, will decrease.

(2) When solid NH_4Cl is added to NH_4OH solution, the ⇌brm

$$NH_4OH \rightleftharpoons NH_4^+ + OH^-$$

shifts to the left. Thereby the ⇌brm concentration of OH^- decreases. This procedure of reducing the concentration of OH^- ions is used in qualitative analysis.

Example: find the degree of dissociation of HF in 1M aqueous solution. The value of K for the ionic ~~⇌~~brm $HF \rightleftharpoons H^+ + F^-$ is 7.2×10^{-4}.

Solution:

HF ~~dissociates~~ in Water to form H^+ and F^- ions. On reaching ⇌brm

$$HF \rightleftharpoons H^+ + F^-$$

Thus 1 mole of HF taken initially dissociate to yield 1 mole of H^+ and 1 mole of F^-.

If x be the degree of dissociation, the Concⁿ terms at ⇌brm are:-

$$[HF] = 1 - x \text{ mol}L^{-1}$$

$$[H^+] = x \text{ mol}L^{-1}$$

$$[F^-] = x \text{ mol}L^{-1}$$

On substituting the values to the above expression,

$$K = 7.2\times10^{-4} = \frac{[H^+][F^-]}{[HF]} = \frac{(x)(x)}{1.00 - x}$$

If $x \ll 1$, we Can write:-

$$7.2\times10^{-4} = \frac{x^2}{1.00}$$

$$\therefore x^2 = 7.2\times10^{-4}$$

$$x = \sqrt{7.2\times10^{-4}} = \underline{2.7\times10^{-2}}$$

The degree of dissociation of HF in 1M solution is $\underline{\underline{2.7\times10^{-2}}}$

Factors that influence the Degree of Dissociation :-

(1) Nature of Solute (2) Nature of the Solvent (3) Concentration (4) Temperature.

SOLUBILITY EQUILIBRIA AND THE SOLUBILITY PRODUCT

When an ionic solid substance dissolves in water, it dissociates to give separate cations and anions. As the concentration of the ions in solution decreases, they collide and reform the solid phase. Ultimately, a dynamic equilibrium is established b/w the solid phase and the cations and anions in solution.

For instant, the equilibrium b/w the sparingly soluble solid ionic silver chloride and its ions in the solution is represented as:-

$$AgCl_{(s)} \rightleftharpoons AgCl_{(aq)} \rightleftharpoons Ag^+ + Cl^-.$$

Applying the law of Mass Action, the equilibrium constant is given by:

$$K = \frac{[Ag^+][Cl^-]}{[AgCl]} \quad \text{— — — —} \quad (1)$$

But, since the unionised AgCl is in contact with the solid AgCl whose active mass is indeed constant, the concentration (more correctly activity) of unionised AgCl can be taken constant. Therefore eqn (1) reduces to

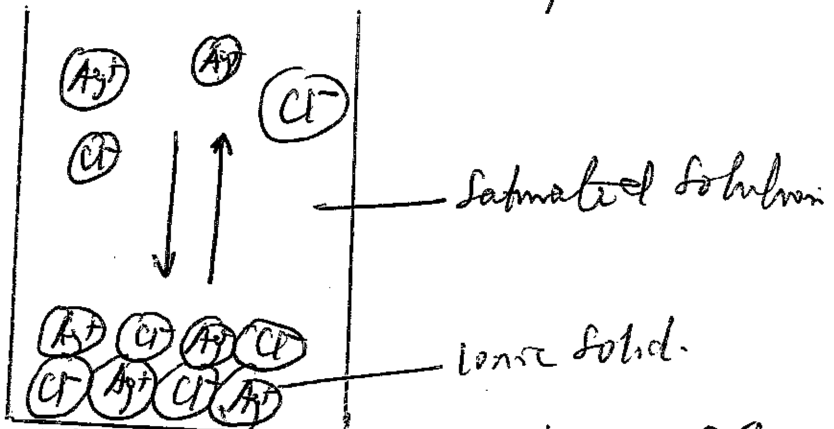
$$K_{sp} = [Ag^+][Cl^-] \quad \text{— —} \quad (2)$$

The constant K_{sp} is referred to as the solubility product constant or simply the solubility product of AgCl. In general, for any sparingly soluble electrolyte represented by AB, in contact with its saturated solution, the equilibrium is represented by:-

$$AB_{(s)} \rightleftharpoons AB_{(aq)} \rightleftharpoons A^+ + B^-$$

and the solubility product is $K_{sp} = [A^+][B^-]$.

A saturated solution is a solution in which the dissolved and undissolved solute are in equilibrium: eg.



Equilibrium b/w an ionic solid and ions in saturated solution

A saturated solution represent the limit of a solute's ability to dissolved ~~and undissolved solute are in~~ in a given solvent. This is a measure of the "solubility" of the solute:

The solubility (S) of a substance in a solvent is the concentration in the saturated solution. ~~the~~ Solubility of solute may be represented in grams per 100mL of solution. It can also be expressed in moles per litre.

Molar solubility is defined as the number of moles of the substance per one litre of the solution

Example: Calculate the solubility of $NiCO_3$ in moles per litre and grams per litre. The value of K_{sp} for $NiCO_3 = 1.4\times10^{-7}$.

Solution:

$$NiCO_{3\,(s)} \rightleftharpoons Ni^{2+} + CO_3^{2-}$$

$$\therefore K_{sp} = [Ni^{2+}][CO_3^{2-}]$$

$$1.4\times10^{-7} = [Ni^{2+}][CO_3^{2-}].$$

from the above ≡brm reaction, one mole of $NiCO_3$ dissociates to produce one mole of Ni^{2+} and one mole of CO_3^{2-}. If S molL^{-1} be the solubility of $NiCO_3$, the ≡brm Concn of Ni^{2+} is S molL^{-1} and that of CO_3^{2-} is S molL^{-1}. On substituting the values in Ksp, we have,

$$1.4\times10^{-7} = [S\,molL^{-1}][S\,molL^{-1}]$$

$$S^2 = 1.4\times10^{-7}$$

$$S = 3.7\times10^{-4}\,molL^{-1}.$$

This means that the solubility of $NiCO_3$ is 3.7×10^{-4} molL^{-1}.

Therefore, the solubility in g/L is $= 3.7\times10^{-4}\,molL^{-1}\times118.7\,g/mol$

$= \underline{0.044\,g/L}$.

__NB__: When two reacting solutions are mixed, Cal. the Concn of each ion in the solution in which pptn is produced. The ionic product Q, is then Calculated. We know that Ksp is the ionic product when the ions are in contact with the solid substance. Therefore, the pptn will occur for any higher ion Concn. If $Q > K_{sp}$ pptn will occur

$Q < K_{sp}$ no pptn will occur.

__Example__: A 200ml of $1.3\times10^{-3}M$ $AgNO_3$ is mixed with 100ml of $4.5\times10^{-5}M$ Na_2S solution. Will pptn occur? ($K_{sp} = 1.6\times10^{-49}$).

(iii)

Solution:

The rxn that would cause pptn is

$$2Ag^+ + S^{2-} \rightleftharpoons Ag_2S_{(s)}$$

The ionic product $Q = [Ag^+]^2[S^{2-}]$

∴ the molar concentrations of Ag^+ and S^{2-} ions:-

$$[Ag^+] = 1.3 \times 10^{-3} M \times \frac{200ml}{(200+100)ml} = 8.7 \times 10^{-4} M$$

$$[S^{2-}] = 4.5 \times 10^{-5} M \times \frac{100ml}{300ml} = 1.5 \times 10^{-5} M$$

$$\therefore Q = (8.7 \times 10^{-4})(4.5 \times 10^{-5}) = 13.1 \times 10^{-9}$$

but $K_{sp} = 1.6 \times 10^{-49}$

∴ $Q > K_{sp}$ and pptn will occur.